

King Fahd University of Petroleum and Minerals
College of Computing and Mathematics
Information and Computer Science Department
**SEC 595: Secure Computation and Post-Quantum
Cryptography**
Term 261



Course Information

- Lectures: Sunday & Tuesday, 20:10-21:25 PM
- Location: Building 24-137

Course Description

Mathematical foundations: algebraic structures, hardness assumptions, and lattice primitives (LWE/RLWE). Lattice-based Post-quantum cryptography: Key encapsulation mechanisms, digital signatures. Homomorphic Encryption: partial (PHE), leveled (LHE), and fully homomorphic encryption (FHE); bootstrapping; BGV, BFV, and CKKS schemes; noise management, packing, and rescaling. Secure computation: secret sharing, oblivious transfer, garbled circuits and Yao's protocol. Zero-knowledge proofs. Applications: privacy-preserving machine learning.

Prerequisites Graduate standing

Course Objectives

The objectives of this course are:

- i. Introduce the students to the emerging field of encrypted computing.
- ii. Expose students to the state-of-the-art algorithms in homomorphic encryption and secure multiparty computation

Learning Outcomes

After taking this course, students will have the ability to:

1. Explain various encrypted computing techniques and algorithms.
2. Identify the advantages and challenges of different encrypted computing algorithms.
3. Apply appropriate encrypted computing techniques based on the application's requirements.
4. Design new encrypted computing techniques and protocols.

Textbook and references

1. Chen K, Yang Q. Privacy-Preserving Computing: For Big Data Analytics and AI. Cambridge University Press; 2023.
2. Dan Boneh and Victor Shoup, A Graduate Course in Applied Cryptography. ([link](#))
3. A Pragmatic Introduction to Secure Multi-Party Computation ([link](#))
4. Katz J, Lindell Y. Introduction to Modern Cryptography. 3rd ed. CRC Press; 2021.
5. Bernstein DJ, Buchmann J, Dahmen E, editors. Post-Quantum Cryptography. Springer; 2009.
6. Cramer R, Damgård I, Nielsen JB. Secure Multiparty Computation and Secret Sharing. Cambridge University Press; 2015.
7. Thaler J. Proofs, Arguments, and Zero-Knowledge. Foundations and Trends in Privacy and Security; 2022. <https://people.cs.georgetown.edu/jthaler/ProofsArgsAndZK.pdf>
8. Dwork C, Roth A. The Algorithmic Foundations of Differential Privacy. 2014. <https://www.cis.upenn.edu/~aaron/privacybook.pdf>
9. List of papers
 - a. Computing Arbitrary Functions of Encrypted Data <https://people.csail.mit.edu/vinodv/6892-Fall2013/GentryCACM.pdf>
 - b. On Lattices, Learning with Errors, Random Linear Codes, and Cryptography <https://people.csail.mit.edu/vinodv/6892-Fall2013/regev.pdf>
 - c. The Learning with Errors Problem <https://people.csail.mit.edu/vinodv/6892-Fall2013/lwesurvey.pdf>
 - d. Homomorphic Encryption for Arithmetic of Approximate Numbers <https://eprint.iacr.org/2016/421.pdf>
 - e. Craig Gentry's thesis <https://crypto.stanford.edu/craig/craig-thesis.pdf>
 - f. (Leveled) Fully Homomorphic Encryption without Bootstrapping <https://people.csail.mit.edu/vinodv/6892-Fall2013/BGV.pdf>
 - g. https://files.boazbarak.org/crypto/lec_15_FHE.pdf
 - h. <https://securecomputation.org/docs/pragmaticmpc.pdf>
 - i. On Ideal Lattices and Learning with Errors over Rings (RLWE) <https://eprint.iacr.org/2012/230.pdf>
 - j. CRYSTALS-Kyber: A CCA-Secure Module-Lattice-Based KEM <https://eprint.iacr.org/2017/634.pdf>
 - k. CRYSTALS-Dilithium: A Lattice-Based Digital Signature Scheme <https://eprint.iacr.org/2017/633.pdf>
 - l. Falcon: Fast-Fourier Lattice-Based Compact Signatures over NTRU <https://falcon-sign.info/falcon.pdf>
 - m. NIST FIPS 203/204/205 – PQC standards <https://csrc.nist.gov/projects/post-quantum-cryptography>
 - n. Shamir A. How to Share a Secret. Communications of the ACM; 1979. <https://dl.acm.org/doi/10.1145/359168.359176>
 - o. Foundations of Garbled Circuits <https://eprint.iacr.org/2012/265.pdf>
 - p. Extending Oblivious Transfers Efficiently (IKNP OT extension) <https://www.iacr.org/archive/crypto2003/27290145/27290145.pdf>
 - q. Groth J. On the Size of Pairing-Based Non-Interactive Arguments (Groth16) <https://eprint.iacr.org/2016/260.pdf>
 - r. Fast Reed-Solomon Interactive Oracle Proofs of Proximity / STARKs <https://eprint.iacr.org/2018/046.pdf>
 - s. PLONK: Permutations over Lagrange-bases for Oecumenical Noninteractive Arguments of Knowledge <https://eprint.iacr.org/2019/953.pdf>
 - t. GAZELLE: A Low Latency Framework for Secure Neural Network Inference <https://eprint.iacr.org/2018/073.pdf>

Useful Links (Libraries and curated lists)

1. OpenMined <https://blog.openmined.org/private-machine-learning-explained/>
2. Awesome HE <https://github.com/jonaschn/awesome-he>
3. HELib library <https://github.com/shaih/HELlib>
4. Microsoft SEAL <https://www.microsoft.com/en-us/research/project/microsoft-seal/>
5. MIT HE List <https://people.csail.mit.edu/vinodv/FHE/FHE-refs.html>
6. OpenFHE Community <https://www.openfhe.org/>
7. Lattigo (Go lattice-based HE library) <https://github.com/tuneinsight/lattigo>
8. TFHE-rs (Zama, Boolean/integer FHE) <https://github.com/zama-ai/tfhe-rs>
9. Concrete ML (FHE for machine learning) <https://github.com/zama-ai/concrete-ml>
10. HomomorphicEncryption.org standardization consortium <https://homomorphicencryption.org/>
11. MP-SPDZ (multi-protocol MPC framework) <https://github.com/data61/MP-SPDZ>
12. EMP-toolkit (garbled circuits and OT) <https://github.com/emp-toolkit>
13. Awesome MPC (curated list) <https://github.com/rdragos/awesome-mpc>
14. Circom (ZK circuit compiler) <https://docs.circom.io/>
15. halo2 (Zcash proving system) <https://zcash.github.io/halo2/>
16. gnark (Consensys ZK-SNARK library in Go) <https://github.com/Consensys/gnark>
17. Awesome Zero-Knowledge Proofs (curated list) <https://github.com/matter-labs/awesome-zero-knowledge-proofs>
18. ZKProof standards community <https://zkproof.org/>
19. NIST Post-Quantum Cryptography project <https://csrc.nist.gov/projects/post-quantum-cryptography>
20. Open Quantum Safe / liboqs (PQC implementations) <https://openquantumsafe.org/>
21. OpenDP (differential privacy tooling) <https://opendp.org/>

Evaluation

Paper Presentation	10%
Project	50%
Midterm Exam	20%
Final Exam	20%

List of Topics

Week	Topic	Key Content
1	Foundations & Mathematical Review	Modern algebraic structures, hardness assumptions, lattice primitives (LWE/RLWE)
2	Post-Quantum Cryptography I	PQC paradigms, code-based cryptography, lattice-based key encapsulation mechanisms (KEMs)
3	Post-Quantum Cryptography II	Digital signatures (Dilithium/Falcon), NIST PQC standardization, migration strategies
4	Fundamentals of Homomorphic Encryption	PHE, LHE, and FHE concepts; Gentry's bootstrapping principle

5	FHE Schemes for Exact Computation	BGV and BFV schemes, noise management, ciphertext packing techniques
6	FHE for Approximate Computation	The CKKS scheme, fixed-point arithmetic, rescaling
7	Paper Presentations	
8	Secret Sharing & Oblivious Transfer	Shamir's Secret Sharing, additive secret sharing, OT protocols (1-out-of-2 OT, OT extension)
9	Garbled Circuits & Yao's Protocol	Boolean circuit representations, garbled gate evaluation, point-and-permute optimizations
10	General Secure Multi-Party Computation (MPC)	BGW/GMW protocols, malicious vs. semi-honest security models, threshold cryptosystems
11	Introduction to Zero-Knowledge Proofs (ZKP)	Interactive proofs, Sigma protocols, commitment schemes, Fiat-Shamir heuristic
12	Non-Interactive ZKPs (zk-SNARKs & zk-STARKs)	Arithmetic circuits, R1CS/QAP formulations, trusted setup vs. transparent setup
13	Practical ZKP Systems	Proof generation, verification efficiency, modern tooling (Circom, halo2, gnark)
14	Privacy-Preserving Machine Learning (PPML)	Encrypted inference and training combining HE, MPC, and ZKPs
15	Project Presentation	

Course Policies

- **Course Website & Participation:** Students are required to periodically check the course website on Blackboard the MS Teams.
- **Attendance:** Regular attendance is a university requirement; hence attendance will be taken at the beginning of each lecture.
 - Missing more than **6 lectures** will result in a **DN grade without warning**.
 - Official excuses must be presented to the instructor no later than one week of returning to classes.
- **Re-grading policy:** If you have a complaint about any of your grades, discuss it with the instructor no later than a week from distributing the grades (except for the final). Only legitimate concerns on grading should be discussed.
- **Office Hours:** Students are encouraged to visit faculty in the office hours to clarify any part of the material that is not clear.
- **Academic honesty:** Students are expected to abide by all the university regulations on academic honesty. **Cheating will be reported to the Department Chairman** and will be severely penalized. Although collaboration and sharing knowledge is highly encouraged, copying others' work (classmates, others or from the web) without proper citation, either in part or full, is considered plagiarism. Whenever in doubt, review the university guidelines or consult the instructor. **Cheating in whatever form will result in an F grade.**
- Absolute academic integrity is expected of every KFUPM student in all academic undertakings. Most essential values academia are grounded on the concept of honesty with respect to the intellectual efforts of oneself and others. While both students and faculty of KFUPM assume the responsibility of maintaining and furthering these values, this memo is concerned specifically with the conduct of students. The KFUPM have a zero-tolerance policy towards cheating, plagiarism or any other violation to the Code of Honesty.
 - Examples of Violations of Code of Academic Honesty
 - Knowingly representing the work of others as one's own.
 - Using, obtaining, or providing unauthorized assistance on examinations, quizzes, assignment, or any other academic work.
 - Fabricating results in support of laboratory or field work.
 - Examinations.
 - No student may take an examination for another student. The student is responsible for understanding the conditions and rules under which the examination will be taken.
 - Dishonesty or cheating in examinations which is defined as the use of inappropriate or unauthorized materials, information, or study aids in an exam. Unless the instructor directs otherwise, an examination is assumed to be solely a student's own work.
 - No communication is allowed among students either through voice, written, electronic, or any other form of transmission, nor are students permitted to consult books, papers, study aids or notes without explicit permission of the course instructor.
 - Violating exam rules and regulations by bringing smart electronic devices (such as mobile phones, smart watches, earbuds, etc.) to the exam venue is considered a cheating attempt.

Comments from previous offering

- Better course organization with more organized slides and teaching materials even though the handwriting notes
- Was helpful
- Better form of course material (e.g., in ppt slides rather than hand written)
- I enjoyed it all ... may be we can increase the content or have the lecture done through ppt instead of note files although the notes were nice as the instructor was going step by step and not leaving anything behind.
- I suggest giving exercises after completing each topic to enable the student to practice and understand the nature of exam questions in advance.
- I think everything is amazing about the course.
- The course is well structured I dont find any improvement to suggest for this course at the moment